

# **SCHOOL MANAGEMENT SYSTEM FOR MATUNDU PRIMARY SCHOOL**

## **SYSTEM DESIGN SPECIFICATION**

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# 1. SYSTEM ARCHITECTURE

## 1.1 Three-Tier Architecture

The system uses a three-tier client-server architecture:

| Tier     | Technology                     | Purpose                    |
|----------|--------------------------------|----------------------------|
| Client   | React 18 + Vite + Tailwind CSS | Single Page Application UI |
| Server   | Node.js 20 + Express.js        | RESTful API backend        |
| Database | PostgreSQL 15 + Prisma ORM     | Relational data storage    |

## 1.2 Deployment Architecture

Frontend: Cloudflare Pages (auto-deploy from GitHub)

Backend: Render (auto-deploy from GitHub, Port 3000)

Database: Supabase (Managed PostgreSQL)

## 2. DATABASE DESIGN

### 2.1 Entity-Relationship Overview

The database contains 23+ tables with UUID primary keys. Key relationships:

- User → Student, Staff, Guardian (one-to-one)
- Student → Class (many-to-one), Attendances, Scores, Payments (one-to-many)
- Class → Grade, Stream, AcademicYear (many-to-one)
- Assessment → Subject, Term (many-to-one), Scores (one-to-many)
- StudentInvoice → Payments, InvoiceItems (one-to-many)

### 2.2 Key Tables

| Table           | Purpose                 | Key Fields                                                 |
|-----------------|-------------------------|------------------------------------------------------------|
| User            | Authentication accounts | id, email, password, role                                  |
| Student         | Learner profiles        | admissionNumber,<br>firstName, classId,<br>admissionStatus |
| Staff           | Teacher/staff profiles  | employeeNumber,<br>firstName, qualification                |
| Class           | Grade+Stream+Year combo | name, gradeId, streamId,<br>academicYearId                 |
| Attendance      | Daily records           | date, status, studentId,<br>classId                        |
| Assessment      | Test/exam definitions   | name, type, maxScore,<br>weight, subjectId                 |
| AssessmentScore | Student scores          | score, grade, studentId,<br>assessmentId                   |
| ReportCard      | Termly reports          | totalScore, averageScore,<br>rank                          |
| FeeStructure    | Fee configuration       | name, amount, gradeId,<br>termId                           |
| StudentInvoice  | Billing                 | invoiceNo, totalAmount,                                    |

|                |              |                                           |
|----------------|--------------|-------------------------------------------|
|                |              | paidAmount, balance                       |
| <b>Payment</b> | Transactions | amount, method, status,<br>transactionRef |

### 3. MODULE DESIGN

#### 3.1 Backend Feature Module Pattern

Each feature follows a three-file pattern: routes.js (route definitions + validation), controller.js (request handling), service.js (business logic + DB queries).

Feature modules: auth, students, staff, academic, attendance, assessments, reports, fees, communication, dashboard.

#### 3.2 Middleware Pipeline

| Order | Middleware   | Purpose                 |
|-------|--------------|-------------------------|
| 1     | CORS         | Cross-origin handling   |
| 2     | Helmet       | Security headers        |
| 3     | Rate Limiter | Request throttling      |
| 4     | Request ID   | Unique request tracking |
| 5     | Body Parser  | JSON parsing            |
| 6     | Sanitize     | Input sanitization      |
| 7     | Auth (JWT)   | Token verification      |
| 8     | RBAC         | Role permission check   |
| 9     | Validate     | Request validation      |

#### 3.3 Authentication Flow

1. Client sends POST /auth/login with email and password.
2. Server validates credentials with bcrypt.compare().
3. Server generates JWT access token (15 min) and refresh token (7 days).
4. Client stores tokens and includes access token in Authorization header.
5. Server verifies JWT and checks RBAC permissions on each request.

## 4. FRONTEND DESIGN

### 4.1 Technology Stack

| Technology   | Purpose             |
|--------------|---------------------|
| React 18     | Component-based UI  |
| Vite         | Build tool          |
| Tailwind CSS | Styling             |
| Zustand      | State management    |
| Axios        | HTTP client         |
| React Router | Client-side routing |
| Recharts     | Data visualization  |
| Lucide React | Icons               |

### 4.2 Page Structure (17+ Pages)

- Login – Authentication form
- Dashboard – Role-specific analytics
- Students – Student CRUD with search/filter
- Admissions – Approval/rejection workflow
- Staff – Staff management
- Classes – Grade/stream management
- Attendance – Daily marking interface
- Assessments – Score entry
- Reports – Report card generation
- Fees – Financial management
- Timetable – Weekly grid view
- Announcements – Communication
- Settings – System configuration
- Profile – User profile
- System Status – Health monitoring

## 5. SECURITY DESIGN

| Aspect           | Implementation                                  |
|------------------|-------------------------------------------------|
| Authentication   | JWT (HS256), 15min access + 7day refresh tokens |
| Password Hashing | bcrypt with 12 salt rounds                      |
| Authorization    | RBAC middleware per route                       |
| Input Validation | express-validator on all fields                 |
| Sanitization     | Custom middleware strips HTML/scripts           |
| Rate Limiting    | Redis-backed request limits per IP              |
| Security Headers | Helmet.js (XSS, CSRF, clickjacking)             |
| Audit Trail      | All critical operations logged                  |